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Research paper

The impacts of return policies and perception heterogeneity on sales platforms for experience goods: An experimental study

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ABSTRACT

On sales platforms for experience goods, consumers only know the distribution of a good's value before purchase, while its actual value is revealed after the purchase. However, consumers may perceive the distribution's range differently, and the presence of a return policy can also influence strategic interactions in the market. Utilizing a post-price framework and controlled laboratory experiments, this paper provides the first investigation of experience goods markets that directly examines (1) the presence of a return policy and (2) the level of consumers' perception heterogeneity about the range of the value's corresponding distribution. We find that return policies result in lower prices in markets for experience goods. In addition, a larger perception heterogeneity reduces prices and increases return rates. Our results highlight potential downsides of mandated return policies on online sales platforms, as they reduce sellers' profits without improving consumer welfare.

1. Introduction

Among the various markets adopting online sales platforms, the market for experience goods stands out as an important component of the digital economy. An experience good is a product or service whose true value can only be determined with certainty after it has been purchased (Nelson, 1970; Che, 1996). Before making a purchase, consumers only have a rough estimate of the good's value. The true value of the good, which is drawn from a random distribution, becomes apparent only after the purchase.

Understanding behavior in such markets requires attention to two distinct forces that may affect how valuation's uncertainty is perceived and resolved by consumers. First, consumers may differ in how they perceive the range of the underlying valuation's distribution, even for the same experience good. When perception heterogeneity is small, consumers share similar perceptions about the distribution's upper and lower bounds; otherwise, their perceived ranges overlap less. Second, mandated return policies play an institutional role in mitigating the uncertainty of the good's valuations. The rapid growth of e-commerce has resulted in higher return rates, prompting corresponding policy changes with ambiguous impacts on online sales platforms.[1] For instance, the recent Amending Directive 2023/2673 of the European Union (EU) enriches the European Directive 2011/83/EU by allowing consumers to exercise the right of withdrawal for all purchases concluded at a distance.[2] The right is institutionally protected, meaning that sellers are legally obligated to provide the return policy and cannot opt out.

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[1] For example, in the United States, the National Retail Federation estimates that 16.9% of retailers' annual sales in 2024 will be returned by consumers. According to the report of the European Parliamentary Research Service, the return rate of the European Union was 19.2% in 2023.

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In this paper, we examine how return policies and perception heterogeneity may affect behavior in experience goods markets via controlled laboratory experiments. After all, each factor has received limited empirical attention on its own, and their interaction is likely to be even more complex—while return policies are expected to reduce the risk of purchase, the magnitude of this effect may also depend on how consumers perceive potential valuations of the experience good.

In our experiment, a seller (she) and a buyer (he) make strategic decisions about trading a single unit of an experience good. They interact under a post-price sales mechanism that follows the typical steps of online sales platforms.[3] The value of the good is drawn from a random distribution whose range is the buyer's private information. Buyers can be of two types, varying by the range of the distribution. The two ranges must overlap in order to exclude the possibility of price discrimination. The seller first sets a uniform price for the good, without knowing the buyer's exact type. The buyer makes decisions after being informed of his own type and the uniform price. If there is no return policy, the buyer's only decision is about whether to purchase the good or not. If a return policy is in place, the buyer has an additional option to return the good to the seller at a cost after purchase. For both cases, the buyer learns the true value of the good only after purchase.

The primary purpose of the experiment is to isolate direct effects of two distinct channels: return policy and perception heterogeneity. Moreover, there is a lack of observational data on consumers' potential valuations of experience goods, not to mention the possibility that these valuations may also be influenced by buyers' biased beliefs about the perceived range of the good's valuation. The key advantage of utilizing laboratory experiments in this study is that the controlled environment allows us to induce the range from which subjects perceive the good's valuation. The laboratory setting offers greater control than observational data, where perceptions of the same experience good tend to be more heterogeneous and are often affected by self-reporting biases.

We adopt a 2×2 design that motivates the examination of not only direct effects from each distinct channel, but also their potential interactions. There are twenty rounds in our experiment, including both within-subject and between-subjects variations. We start by looking at a market of experience goods without a return policy in the first ten rounds. Then, we introduce the return policy to the market in the next ten rounds. This within-subject variation aims to examine the effect of policies that guarantee buyers' withdrawal rights from the institutional perspective. Meanwhile, our between-subjects variation focuses on different levels of perception heterogeneity. In the treatment where the perception heterogeneity is relatively small, the ranges of the two distributions overlap significantly. In contrast, when perception heterogeneity is large, the two ranges overlap, but the extent of overlap is to a lesser extent. This treatment examines experience goods characterized by relatively large heterogeneity in buyers' perceptions of the good's potential valuation range.

Our hypotheses are constructed to examine interesting cases where the return policy may bring negative impacts to the market, especially for sellers. We hypothesize that under the "Small Difference" (resp. "Large Difference") treatment, the return policy will make the seller's price decrease (resp. increase). Overall, however, the presence of a return policy and greater perceived heterogeneity each contribute to lowering the seller's price. We also hypothesize that purchase rates are identical across both within-subject and between-subject variations because the seller's price ensures that buyers with higher valuation ranges always purchase, while others are indifferent between purchasing and not. However, the "Large Difference" treatment generates higher return rates than the "Small Difference" treatment. Eventually, both the presence of a return policy and a large difference across consumers' perception heterogeneity make the seller worse off.

Based on our experimental data, we find that the presence of a return policy reduces the seller's price. Interestingly, the "Large Difference" treatment, rather than the "Small Difference" treatment, becomes the main driving force of the price decrease, even though prices are predicted to increase due to the presence of a return policy when the perception heterogeneity is large. The underlying mechanism of this equilibrium deviation is that sellers target high-type buyers when the perception gap is large and there is no return policy in the market. Eventually, the return policy hurts the seller but has a minimal impact on the buyer's welfare level. These observations reveal the negative effects of mandated return policies in online sales platforms for experience goods.

Meanwhile, how will the degree of perception heterogeneity affect market behavior? Our results show that the seller's price is lower when the perception heterogeneity among buyers is relatively larger. The large gap between buyers' perception heterogeneity reduces the purchase rate but increases the return rate, even after controlling for the effects driven by price changes. Consequently, when buyers hold more heterogeneous perceptions of the good's valuation, the treatment effects suggest that this gap does not benefit the buyer but instead reduces the seller's welfare.

The policy implication of our paper is on the potential downsides of implementing mandatory return policies, especially from the seller's perspective. We provide first-hand experimental evidence that the presence of a return policy is likely to hurt sellers

[2] See https://europa.eu/youreurope/citizens/consumers/shopping/guarantees-returns and https://eur-lex.europa.eu/eli/dir/2023/2673/oj/eng. If a consumer purchases a product or service online, by phone or from a seller at his doorstep, he can always cancel it within fourteen days without providing any justification. In numerous developing regions, returns are either absent or apply only under restrictive conditions. Here, the return policy is mandated, rather than an endogenous choice by sellers.

[3] Here, we interpret an online sales mechanism with a return policy as a problem in which the buyer learns additional private information about his valuation once he has purchased the good. The reason is that the optimal contract of a dynamic screening problem with costly withdrawal rights coincides with the one in a static screening problem, which is a simple post-price sales mechanism (Krähmer and Strausz, 2015). Our concentration on post-price sales mechanisms also keeps the simplicity of the experiment without affecting players' strategic integrations, which are built on the foundations of both static screening mechanisms (Baron and Myerson, 1982) and dynamic screening mechanisms (Courty and Li, 2000). However, dynamic screening mechanisms require subjects' full comprehension of complex sales rules. Even in simplified mechanism design settings — such as eliciting individual valuations — comparisons between simple and dynamic mechanisms have produced mixed results (Brown et al., 2025).

but does not benefit buyers in the context of experience goods. While it is crucial for policymakers to emphasize the benefits of guaranteeing buyers' withdrawal rights, mandating a return policy may not always be optimal. Allowing sellers and buyers to decide on the leniency of return policies may have the potential to outperform mandated return policies, at least in markets for experience goods.

The remainder of the paper is structured as follows. Section 2 reviews related literature. Section 3 describes the experimental design and procedures. Section 4 states our testable hypotheses. Section 5 discusses our experimental results. Section 6 concludes.

2. Related literature

This paper contributes to two strands of literature. The first is about the flourishing literature related to return policies. Previous literature has documented that providing full refunds may not be optimal for sellers in non-strategic environments (Pasternack, 1985; Chang and Pao, 2007; Janakiraman et al., 2016). However, the total effect in strategic environments is mixed due to risk aversion (Che, 1996), high return rates (Davis et al., 1998) and the abuse of return policies (Reynolds and Harris, 2009).

As for buyers, the main focus of empirical literature is on the signaling effect of return policies (Davis and Holt, 1994; Marvel and Peck, 1995; Su, 2009; Wang, 2009; Becher and Zarsky, 2010). The intuition is that providing a lenient return policy can signal high quality of the good and therefore should increase both the good's price and purchasing tendency (Moorthy and Srinivasan, 1995).[4] In contrast, Cui et al. (2020)'s simulation shows that a return policy can hurt sellers. We focus on a screening mechanism grounded in prior theoretical work with the key assumption that the seller is uncertain about the valuation's exact range (Che, 1996; Krähmer and Strausz, 2015; Becher et al., 2023). Our study also provides a novel channel of explanation, which is the perception heterogeneity of experience goods.

Several theories have examined the role of return policies in screening mechanisms (Hinnosaar and Kawai, 2020; Zhang et al., 2022; Lyu, 2024),[5] Our setup differs from previous models that assume valuations are drawn from the same distribution as common knowledge (Che, 1996) or signals (Hinnosaar and Kawai, 2020).[6] As for experiments related to return policies, both Wood (2001) and Wang (2009) investigated the role of endowment effects in non-strategic environments. A theory-based laboratory experiment conducted by Cadsby et al. (2016) examined how return policies affect bidding behavior in sealed-bid auctions. Different from those studies, we concentrate on experience goods markets, where it is crucial to assume that buyers' perceptions of the good's potential valuation differ. Our paper provides the first empirical examination of mandated return policies in a strategic platform that incorporates perception heterogeneity among buyers.

This paper also contributes to the growing literature of experience goods markets in terms of both theoretical foundations (Liebeskind and Rumelt, 1989; Kranton, 2003; Huck and Tyran, 2007; Jing, 2016; Chen et al., 2022a,b) and empirical investigations (Ackerberg, 2003; Reinstein and Snyder, 2005; Ali and Nauges, 2007; Weathers et al., 2007; Huang et al., 2009). Besides, several laboratory experiments have examined potential candidates that can affect decisions in experience goods markets, including moral hazard (DeJong et al., 1985; Lynch et al., 1986; Holt and Sherman, 1990), repeated interactions (Davis and Holt, 1994), trust (Huck et al., 2012), transparency (Henze et al., 2015), price competition (Huck et al., 2016), and reputation (Solimine and Isaac, 2023).[7] To the best of our knowledge, no empirical study has yet investigated the impact of mandated return policies on sales platforms for experience goods.

3. Experimental design

The laboratory experiment is framed based on the main features of post-price sales platforms. At the beginning of the experiment, subjects are informed that they are involved in a game played with a pair of two subjects. One subject is randomly assigned as the seller, and the other subject is randomly assigned as the buyer. Each seller is endowed with one unit of experience good.[8] The action of a seller is to set a uniform price for the experience good. After that, the matched buyer chooses whether to purchase the good from the seller or not. Once all subjects have made their decisions, they will be informed of their own payoffs for the round. Subjects will not be informed of other subjects' payoffs.

[4] While return policies may signal product quality in some contexts, such effect is not compatible with our study's setup since our return policy is mandated (Che, 1996; Krähmer and Strausz, 2015). Also, the seller is uncertain about the experience good's true value (so no players know the good's true value at the timing of contracting the terms of trade) and she faces multiple types of buyers.

[5] There is also a growing theoretical literature examining how return policies affect consumers' search behavior (Inderst and Tirosh, 2015; Petrikaitė, 2018; Ibragimov et al., 2024; Janssen and Williams, 2024; Li et al., 2024; Jerath and Ren, 2025).

[6] In practice, return policies are often exogenous and mandated in many regions around the world. For example, in the European Union, the Consumer Rights Directive (2011/83/EU) requires a minimum 14-day right of withdrawal for online and distance purchases, effectively removing firms' discretion over basic return terms. Similar mandated return policies are effective in the UK (Consumer Contracts Regulations 2013, post-Brexit retention of EU rules, https://www.legislation.gov.uk/uksi/2013/3134/contents). While there is no universal return policy in the United States, online sales platforms like Amazon are still required to clearly disclose their return terms, and federal or state laws may impose additional requirements in specific contexts. Including endogenous choice of return options is left for future experiments. However, that still needs the benchmark case of assuming exogeneity of return options.

[7] Brandts and Potters (2018) reviewed the behavioral and experimental literature of experience goods markets. Our design adopts a simplified version of the market in which a seller only interacts with one buyer.

[8] To avoid framing effects, the experience good is defined neutrally as a "virtual item" to subjects. Instructions and screenshots are available as supplementary materials.

Table 1
Summary of the experiment's treatment variations.

	Small Difference	Large Difference
High Type's Distribution	Uniform[60, 100]	Uniform[60, 100]
Low Type's Distribution	Uniform[55, 95]	Uniform[40, 80]
Rounds 1–10	No Return	With Return
Rounds 11–20	No Return	With Return

All numbers are in points (20 points = $1).

Table 2
Timeline for a round in the experiment.

Timeline for a round when there is no return policy (Rounds 1–10)	
Stage 1	• Subjects are randomly matched in pairs.
Stage 2	• The buyer's type and the experience good's value are randomly determined.
Stage 3	• The seller names the price of the experience good.
Stage 4	• The buyer is informed of his own type.
Stage 5	• The buyer decides whether to purchase the experience good or not, after being • informed of the seller's price.
Stage 6	• Subjects are informed of their own payoffs.
Timeline for a round when there is a return policy (Rounds 11–20)	
Stage 1	• Subjects are randomly matched in pairs.
Stage 2	• The buyer's type and the experience good's value are randomly determined.
Stage 3	• The seller names the price of the experience good.
Stage 4	• The buyer is informed of his own type.
Stage 5	• The buyer decides whether to purchase the experience good or not, after being • informed of the seller's price.
Stage 6	• **If the buyer purchases, he is informed of the experience good's value • and decide whether to return it to the seller or not.**
Stage 7	• Subjects are informed of their own payoffs.

There are a total of twenty rounds in each session. Stranger matching is adopted in our experiment—new pairs of buyers and sellers are randomly formed in each round, but the role of each subject is fixed throughout the entirety of the experiment. Subjects' payoffs are denominated in points, with the exchange rate of 20 points = $1.

Table 1 summarizes the experimental design. Table 2 presents the timeline of a round. The direct comparison between "no return" and "with return" is included as the experiment's within-subject variation. In rounds 1–10, each round ends once the buyer decides whether to purchase the experience good from the seller. Starting in round 11, a return policy is introduced to the market. As before, the buyer learns the true value of the good upon purchase, but now has the option to either keep it or return it at a cost.[9]

Buyers can be one of two types, high-type or low-type, randomly assigned in each round with an equal chance. A different type leads to different possible ranges of the valuation's distribution. The valuation of the experience good is drawn from a uniform distribution with an increment of 1 point. Our between-subjects variations, "Large Difference" and "Small Difference", vary by the range of the random distribution that determines the valuation of the experience good. The distribution range of high-type buyers is the same in both between-subjects treatments—Uniform[60, 100]. Under the "Small Difference" (resp. "Large Difference") treatment, the distribution of a low-type buyer's valuation is Uniform[55, 95] (resp. Uniform[40, 80]). Hence, the treatment of "Small Difference" (resp. "Large Difference") represents the case of a relatively small (resp. large) gap of perception heterogeneity among buyers.

All players are endowed with 60 points ($3) at the beginning of each round.[10] The seller's payoff depends on her price and whether the buyer purchases the experience good or not. If the buyer makes a purchase, the seller receives the price of the experience good but also incurs a fixed sales cost of 25 points ($1.25). Thus, the seller can set a uniform price (must be an integer) that is between 25 ($1.25) and 100 points ($5). In rounds 1–10, the buyer's payment depends on the seller's price, the buyer's purchasing decision, and the value of the good. In rounds 11–20, the buyer's payoff is also related to her decision to return the good after learning the true value. The buyer incurs a cost of 5 points ($0.25) if he chooses to return the good.[11]

Instructions of rounds 1–10 were distributed and read aloud to all subjects before the experiment began. Instructions of rounds 11–20 were distributed and read aloud to all subjects before round 11 began. After all rounds, subjects filled out an exit questionnaire

[9] Given the complexity of the screening mechanism for subjects, we let them play the game for ten rounds with fixed roles in each condition related to return policies, rather than randomizing the order for "no return" and "with return" variation for each round of the game. Also, our design does not include a within-subject reversal—i.e., starting with a return policy and then removing it. To the best of our knowledge, such policy reversals are not present in real-world markets. We focus instead on the effects of introducing a return policy, as mandated prohibitions of return policies are uncommon even in traditional sales contexts.

[10] The purpose is to eliminate the possibility of negative payoffs in a round, especially for buyers. Here is an example that happened once in round 1 of the experiment. A buyer mistakenly purchased the good at a price of 100 points but the drawn value was 62 points, With no endowment, he will lose 38 points in the round. The worst-case scenario is that the buyer will lose 60 points (when the price is 100 points but the buyer's valuation is 40 points), hence the 60-point endowment. That also means, for rounds 1–10, it is likely that a buyer makes active decisions by purchasing the item, but the payoff is less than $3 in a round.

consisting of their descriptions of the strategy used in the experiment, a non-incentivized risk-preference task similar to Eckel and Grossman (2008), a non-incentivized ambiguity-preference test with two Ellsberg (1961) questions, and three questions from the Cognitive Reflection Test (CRT) (Frederick, 2005). One round of rounds 1–10 and one round of rounds 11–20 were randomly selected, and subjects were paid based on total payoffs in these two rounds, in addition to a $10 participation payment. The average payment was $16.82.

Subjects for this experiment were 102 undergraduate students at Texas A&M University, recruited using the website, a server based on ORSEE (Greiner, 2015). As the experiment contained between-subjects treatment variations, each subject participated in only one session. Forty-eight subjects participated in the "Small Difference" treatment, while fifty-four subjects participated in the "Large Difference" treatment. Subjects made decisions using z-Tree software (Fischbacher, 2007). Ten in-person sessions were held at the Experimental Research Laboratory at Texas A&M University during spring 2023. Each session took less than 90 min, including instructions, comprehensive quiz questions, and payment procedures.[12]

4. Hypotheses

Our testable hypotheses are derived from a conceptual framework that examines strategic interactions between a seller and a buyer in a post-price sales mechanism. The seller has a single unit of an experience good for sale. The seller sets a uniform price of the experience good at $p > c$. If the good is sold, the seller earns the price p from the buyer but also incurs a sales cost c. If a purchase eventually occurs, the seller's payoff is payments received minus the sales costs, and the buyer's payoff is valuation minus payments if the buyer eventually purchases the good from the seller. When there is a return policy, the buyer incurs a return cost k if he chooses to return the good after purchase.[13] The seller earns zero if the good is not sold at the end.[14] Both players are assumed to be risk-neutral and have quasi-linear utility functions.[15]

Parameters of this experiment match such interesting cases predicted by the framework: Overall, the return policy results in a lower uniform price set by the seller; however, it reduces the price under the "Small Difference" treatment while increasing the price under the "Large Difference" treatment.[16] Hypotheses 1 and 2 are constructed to compare the seller's price in different environments—the seller's uniform price decreases (resp. increases) under the "Small Difference" (resp. "Large Difference") treatment when a return policy is introduced to the market. Based on our parameters, when there is no return policy (so rounds 1–10), the optimal price is 75 (resp. 60) points for the "Small Difference" (resp. "Large Difference") treatment. The reason is that if there is no return policy, the seller has an incentive to set the uniform price at either the expectation of low-type valuations or the lower bound of the high-type's distribution, whichever is higher. When there is a return policy instead, the optimal price is 65 points for both treatments.

When there is a return policy, the seller's pricing behavior depends on the trade-off between the two competing forces: (1) the potential risk of making zero profit due to buyers' returns and (2) the potential additional profit from charging a higher price than the optimal price without a return policy. Under the "Small Difference" treatment, the expected cost of returns outweighs the benefit of charging higher prices to buyers, prompting sellers to reduce the uniform price. In contrast, the "Large Difference" treatment makes the potential gain from higher prices dominate the potential risk of earning zero due to buyers' return behavior. Thus, the direction of price change is different across between-subjects variations, but the average price is higher under the "Small Difference" treatment.

Hypothesis 1. The uniform price is lower in rounds 11–20 than in rounds 1–10. For the "Small Difference" (resp. "Large Difference") treatment, the uniform price is lower (resp. higher) in rounds 11–20 than in rounds 1–10.

Hypothesis 2. The uniform price is higher under the "Small Difference" treatment than the "Large Difference" treatment.

Hypotheses 3–5 investigate how buyers' decisions will be affected by our treatment variations. For buyers, we look at three outcome variables: purchase rate, real purchase rate (the percentage of buyers who purchase the good and keep it at the end), and return rate.

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[11] Therefore, a buyer's payoff is likely to be lower than the guaranteed amount (so $3) if she takes very active actions, especially for rounds 11–20 (when there is a return policy). Consider the case of round 11–20, so when there is a return policy in the market. When buyers choose to purchase the good but also returns the good back to the seller after purchase, she loses 5 points ($0.25) due to the return decision. The buyer's payoff for that round will be $2.75, lower than the payment when she chose not to buy at the beginning of the round ($3). Hence, when subjects' payoff is close to $16, it is not the case that this phenomenon is because of buyers taking no action. Instead, the buyer can be very active by taking two different actions and receive a payoff of $2.75 in a round, lower than the $3 guaranteed amount. If both rounds selected for payment ended in this situation, the buyer's total payoff will be $15.5, lower than $16.

[12] All subjects were required to provide correct answers in all comprehension quiz questions, including quiz questions for rounds 1–10 to start round 1 and quiz questions related to rounds 11–20 to start round 11.

[13] Both c and k are upper-bounded.

[14] Here, the return cost is fully borne by the buyer, while the seller incurs no cost if the good is returned. Our predictions hold when return cost incurred by the seller $c_r \geq 0$ is not sufficiently large. Under the parameters used in this experiment, our predictions still hold when $c_r \leq 20$ points. The profit of any seller is a decreasing function of c_r, further demonstrating the negative effects of return policies to sellers.

[15] See Appendix A for the details of the seller's objective function.

[16] Such circumstances capture the downside of return policies but also generate different directions of return policy's potential effects when perception heterogeneity varies across buyers.

Table 3
Summary statistics for outcome variables of sellers.

Treatment	Rounds	Price mean (std deviation)	Seller's Profit mean (std deviation)
Small Difference (5 sessions, 48 subjects)	1–20	72.685 (8.846)	89.031 (22.719)
	1–10	73.404 (9.406)	91.358 (22.783)
	11–20	71.967 (8.204)	86.704 (22.461)
Large Difference (5 sessions, 54 subjects)	1–20	65.957 (10.854)	83.500 (19.526)
	1–10	67.507 (11.858)	84.485 (20.204)
	11–20	64.407 (9.519)	82.515 (18.810)

For both between-subjects variations, the optimal price is predicted to make high-type buyers purchase for sure, while low-type buyers are indifferent, either with or without a return policy. Thus, purchase rates should be identical across between-subjects treatments. However, the return policy will affect the real purchase rate—93.75% (resp. 75%) under the "Small Difference" (resp. "Large Difference") treatment. That means the probability of return is 6.25% (resp. 25%) under the "Small Difference" (resp. "Large Difference") treatment. As subjects under the "Small Difference" treatment are making decisions related to goods with relatively higher valuations, and since the optimal price is predicted to be the same across between-subjects variations in rounds 11–20 (65 points for both cases), the return rate should also be higher than the "Large Difference" treatment.

Hypothesis 3. For both treatments, there is no significant difference in purchase rate between rounds 1–10 and rounds 11–20, but the real purchase rate is lower in rounds 11–20.

Hypothesis 4. There is no difference between the "Small Difference" and "Large Difference" treatments in terms of purchase rate. However, the real purchase rate is higher under the "Small Difference" treatment than the "Large Difference" treatment.

Hypothesis 5. The return rate is higher under the "Large Difference" treatment.

5. Results

The organization of our empirical analysis is as follows. In Section 5.1, we examine the seller's pricing behavior.[17] In Section 5.2, we extend our analysis to the buyer's behavior. In Section 5.3, we look at the level of players' profits by their roles. For all between-subjects (resp. within-subject) statistical comparisons in the main text, we report p-values from the two-sample t-test (resp. paired t-test).[18] All regressions use standard errors clustered at the subject level. In our regressions, we include subjects' responses to three CRT questions, their risk attitudes, their ambiguity attitudes, and their gender as control variables.[19]

5.1. Seller's behavior

Table 3 provides summary statistics of main outcome variables for sellers. The average price is significantly higher under the "Small Difference" treatment ($p < 0.01$ for both "no return" and "with return" conditions). Meanwhile, the average price is higher in rounds 1–10 ($p \approx 0.06$ under the "Small Difference" treatment while $p \approx 0.01$ under the "Large Difference" treatment).[20] We also performed Ordinary Least Squares (OLS) regressions on the seller's price. Those results are available in Table 4. Supporting the main idea of Hypothesis 1, the presence of a return policy reduces the seller's price. Interestingly, the price decrease is more significant under the "Large Difference" treatment. The uniform price under the "Large Difference" treatment is significantly lower, supporting Hypothesis 2. Perplexingly, the "Large Difference" treatment also reduces the uniform price when there is a return policy in the market.

[17] The analysis of seller's behavior considers only the prices posted by sellers, without accounting for whether goods are returned or not.

[18] We also used the Wilcoxon rank-sum test for the between-subjects comparison and the Wilcoxon matched-pairs signed-ranks test for the within-subject comparison to examine the significance and robustness of our results. All results still hold under those tests.

[19] Summary statistics of those control variables are available in Table A.1. Regression results remain qualitatively unchanged regardless of whether gender is included as a control variable.

[20] Figure A.1 plots the trend of how price evolves across rounds. The "Large Difference" treatment reduces the seller's price by 6.27 points in rounds 1–10 and 8.03 points in rounds 11–20. The presence of a return policy reduces the seller's price by 2.32 points in total. In round 11, the average price dropped to 67.88 points in the "Large Difference" treatment while 73.58 points in the "Small Difference" treatment.

Table 4
OLS regressions on the seller's price.

Panel A: Treatment Effect of Perception Heterogeneity						
	(1) With Return	(2) No Return	(3) Pooled	(4) With Return	(5) No Return	(6) Pooled
Large Difference	−8.022***	−6.238***	−7.130***	−7.559***	−5.897***	−6.728***
	(1.862)	(2.188)	(1.837)	(1.707)	(1.848)	(1.603)
Round	0.174	−0.118	−0.167*	0.174	−0.118	−0.167*
	(0.135)	(0.226)	(0.084)	(0.133)	(0.224)	(0.084)
Constant	67.450***	72.321***	72.667***	69.270***	74.051***	74.442***
	(3.642)	(3.631)	(3.109)	(2.361)	(1.841)	(1.465)
Survey Controls	Yes	Yes	Yes	No	No	No
Observations	510	510	1020	510	510	1020
R^2	0.203	0.110	0.146	0.155	0.071	0.111
Adjusted R^2	0.186	0.090	0.136	0.151	0.067	0.109
Panel B: Treatment Effect of Return Policy						
	(1) Large Difference	(2) Small Difference	(3) Pooled	(4) Large Difference	(5) Small Difference	(6) Pooled
With Return	−3.368*	−1.734	−2.599**	−3.368*	−1.734	−2.599**
	(1.849)	(1.454)	(1.182)	(1.833)	(1.441)	(1.177)
Round	0.027	0.030	0.028	0.027	0.030	0.028
	(0.234)	(0.152)	(0.141)	(0.232)	(0.151)	(0.140)
Constant	64.258***	71.438***	69.150***	67.360***	73.241***	70.127***
	(3.346)	(4.724)	(2.963)	(1.871)	(1.541)	(1.284)
Survey Controls	Yes	Yes	Yes	No	No	No
Observations	540	480	1020	540	480	1020
R^2	0.130	0.152	0.053	0.020	0.007	0.012
Adjusted R^2	0.111	0.132	0.042	0.017	0.003	0.010

Robust standard errors clustered at the subject level in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Recall from Section 4 that sellers are predicted to target both types of buyers: a uniform price secures a purchase from high-type buyers while keeping low-type buyers indifferent. However, when returns are not allowed (rounds 1–10), sellers may instead focus solely on high-type buyers, especially under the "Large Difference" treatment, because they face a trade-off between setting a high price to extract surplus from high-type buyers and lowering the price to capture both types. In this case, the large valuation gap makes excluding low-type buyers attractive for sellers: the lost sales to low-type buyers are outweighed by the higher margins from the profits due to only selling to high-type buyers. On the contrary, when a return policy is in place, setting a high uniform price to exclude low-type buyers becomes less viable due to the risk of returns, which pushes sellers to lower prices and serve both types of buyers. Under the "Small Difference" treatment, the gains from excluding low-type buyers are already limited due to the small gap between different ranges, even without returns—slightly lowering the price captures both types with minimal loss in surplus from high-type buyers. As a result, sellers have little incentive to target only high-type buyers either with or without a return policy under the "Small Difference" treatment.

Fig. 1 shows how sellers' prices in rounds 1–10 relate to subsequent price adjustments in rounds 11–20. The x-axis plots each possible price in rounds 1–10, and the y-axis reports the change in average price in rounds 11–20 relative to the average price in rounds 1–10. Fitted regression lines based on OLS regressions are shown for each treatment. In the "Large Difference" treatment, the optimal price is predicted to increase from 60 to 65 points between the two phases. If sellers target both types of buyers, we should observe clusters of price changes around +5 points, particularly for those who initially priced near 60 points. However, the data reveal considerable heterogeneity, with many sellers under the "Large Difference" treatment either lowering their prices or making only minimal adjustments upwards. [21] This pattern suggests a systematic deviation from the predicted price and instead supports our conjecture that sellers are targeting high-type buyers by setting higher-than-predicted prices in rounds 1–10 under the "Large Difference" treatment.[22]

Based on those results, we find that the presence of a return policy makes the seller reduce her price with the purpose of avoiding potential returns. Besides, perception heterogeneity plays an important role in determining the seller's uniform price—a larger perception heterogeneity increases the seller's likelihood of setting a lower uniform price even with a return policy. With

[21] Figures A.2 and A.3 show that when there is no return policy, the price deviation is higher under the "Large Difference" treatment.

[22] Furthermore, for rounds 1–10 under the "Large Difference" treatment, we calculated the counterfactual of sellers' profits at (1) their observed prices in data and (2) the equilibrium price, given the estimated buyer's purchase behavior with the help of a kernel-weighted local constant estimator separately for high-type and low-type buyers. We found that profits under seller's actual observed prices are higher.

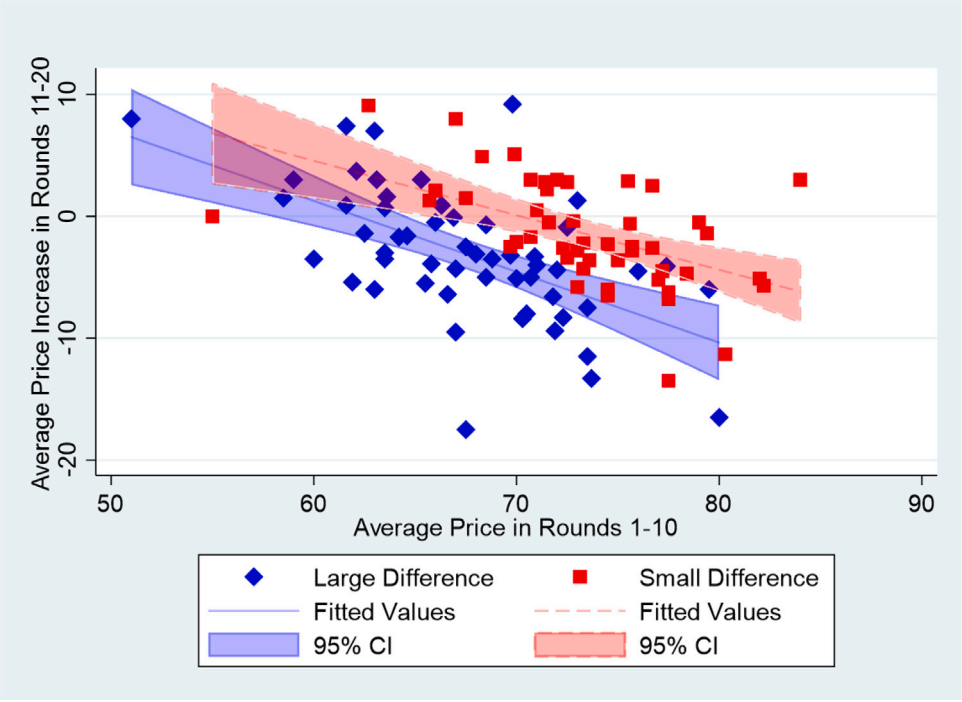


Fig. 1. Observations and fitting curves for price adjustments from "no return" to "with return".

larger perception heterogeneity, sellers are more likely to sell exclusively to high-type buyers in the absence of a return policy, rather than using a uniform price to target both types of buyers. Consequently, it makes the seller worse off due to a lower price when there is a return policy in the market.

Result 1. *The presence of a return policy reduces the seller's uniform price.*

Result 2. *The seller's price is lower under the "Large Difference" treatment than the "Small Difference" treatment.*

5.2. Buyer's behavior

Table 5 provides summary statistics for outcome variables related to buyers. Both the purchase rate and real purchase rate are higher than 60%. To be specific, the purchase rate is roughly 65% for buyers when there is no return policy and roughly 74% when there is a return policy in the market. The high purchase (and real purchase) rates of buyers indicate that even with the guaranteed endowment of 60 points ($3) in each round, buyers are motivated to make active decisions of purchasing based on the incentives provided in our experiment, rather than defaulting to the guaranteed amount. Hence, our experiment's payoff structure is salient, especially to buyers, as all participants are incentivized to act.

We utilize logistic regressions to analyze the buyers' purchasing behavior as buyers' decisions depend on various factors, such as the seller's price and the realized valuation of the good. According to Table 6, we observe clear patterns that support our hypotheses even though sellers' prices are higher than predicted. High-type buyers are more likely to make purchases and real purchases due to their higher ranges of potential valuations. The higher the uniform price is, the smaller the purchase rate and real purchase rate for buyers. The presence of a return policy increases the purchase rate non-significantly but decreases the real purchase rate significantly. The "Large Difference" treatment generates significantly lower real purchase rates than the "Small Difference" treatment.

As for buyer's return decisions, the average return rate is higher in the "Large Difference" treatment ($p < 0.01$). Table 7 presents the results of logistic regressions on the return rate. We also include the seller's price in the regressions as return decisions may also depend on uniform price. While the "Large Difference" treatment increases the return rate, the interaction effect between perception heterogeneity and buyer's type is negative, indicating that the "Large Difference" treatment reduces the return rate for high-type buyers. The return rate decreases as the uniform price increases, which is also reasonable.

One may still wonder whether specific realizations of valuations may affect the real purchase rate and the return rate. We adopted two approaches to address this concern. First, we examined the relationship between drawn valuations and the buyer's decisions related to return in rounds 11–20. According to Fig. 2, the return rate decreases as the drawn valuation increases. On average, the real purchase rate is lower under the "Large Difference" treatment, but higher for valuations that are shared across treatments for

Table 5
Summary statistics for main outcome variables of buyers.

Treatment	Rounds	Purchase Rate mean (std deviation)	Real Purchase Rate mean (std deviation)	Return Rate mean (std deviation)	Buyer's Profit mean (std deviation)
Small Difference (5 sessions, 48 subjects)	1–20	0.713 (0.453)	0.642 (0.480)	0.190 (0.393)	65.550 (11.630)
	1–10	0.679 (0.468)	0.679 (0.468)	N/A N/A	65.238 (12.503)
	11–20	0.746 (0.436)	0.604 (0.490)	0.190 (0.393)	65.863 (10.703)
Large Difference (5 sessions, 54 subjects)	1–20	0.683 (0.466)	0.630 (0.483)	0.148 (0.356)	67.004 (13.839)
	1–10	0.641 (0.481)	0.641 (0.481)	N/A N/A	66.811 (14.410)
	11–20	0.726 (0.447)	0.619 (0.487)	0.148 (0.356)	67.196 (13.268)

Table 6
Logit regressions on buyers' decisions related to purchase.

	(1) Purchase Decision	(2) Real Purchase Decision	(3) Purchase Decision	(4) Real Purchase Decision
High Type	1.902*** (0.240)	1.920*** (0.236)	1.829*** (0.242)	1.862*** (0.239)
With Return	0.250 (0.295)	−0.711*** (0.244)	0.214 (0.287)	−0.723*** (0.236)
Large Difference	−0.939*** (0.260)	−0.802*** (0.235)	−0.953*** (0.267)	−0.871*** (0.253)
Price	−0.143*** (0.019)	−0.141*** (0.018)	−0.140*** (0.019)	−0.139*** (0.018)
Round	−0.005 (0.026)	0.011 (0.023)	−0.003 (0.025)	0.012 (0.023)
Constant	10.879*** (1.425)	10.141*** (1.346)	10.321*** (1.391)	10.089*** (1.320)
Survey Controls	Yes	Yes	No	No
Observations	1020	1020	1020	1020
Log Likelihood	−447.548	−483.788	−457.778	−491.866
Pseudo R^2	0.285	0.277	0.268	0.265

Both dependent variables are binary indicators.
Robust standard errors clustered at the subject level in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

the low-type buyers. This is because the same valuation in that range represents a relatively high valuation for low-type buyers under the "Large Difference" treatment. The high return rate under the "Large Difference" is driven mainly by realized valuations that are not overlapped by the "Small Difference" treatment. But when considering only high-type buyers' choices, the return rate is significantly higher under the "Small Difference" treatment if we restrict our attention to those with drawn valuations that are above 60 points.

We also conducted a simulation exercise in which buyers' behavior was predicted based on sellers' prices. Specifically, a buyer purchases for sure if the price is lower than the expected payoff of the purchase, and he purchases with a 50% chance if the price is equal to the expected payoff of the purchase. The good's valuation is randomly drawn from the distribution that the buyer's type corresponds to if he decides to purchase the good. He will return the good for sure if the price is lower than the valuation plus return cost. If the price is equal to the valuation plus return cost, there is a 50% chance that he will return the good. We repeated the process for 1,000 times to generate simulated data, which we then used to run the same logit regressions as in Tables 6 and 7, and we obtained similar results.[23]

Therefore, we find supportive evidence for Hypotheses 3–5. The presence of a return policy has a minimal impact on the purchase rate but significantly reduces the real purchase rate. Compared with the "Small Difference" treatment, the "Large Difference"

[23] See Tables A.3 and A.4 for the regressions results with the simulation data. The relationship between the realized valuation and buyers' behavior under the simulation data is also plotted in Fig. 2.

Table 7
Logit regressions on buyers' return decisions.

	(1) Return Decision	(2) Return Decision
Large Difference	1.383***	1.752***
	(0.420)	(0.497)
High Type	−0.424	−0.417
	(0.299)	(0.286)
Large Difference × High Type	−2.862***	−2.654***
	(0.713)	(0.671)
Price	0.127***	0.124***
	(0.024)	(0.023)
Round	−0.043	−0.040
	(0.051)	(0.050)
Constant	−8.509***	−9.455***
	(1.918)	(2.059)
Survey Controls	Yes	No
Observations	375	375
Log Likelihood	−131.462	−139.061
Pseudo R^2	0.226	0.181

The dependent variable is a binary indicator.
Robust standard errors clustered at the subject level in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

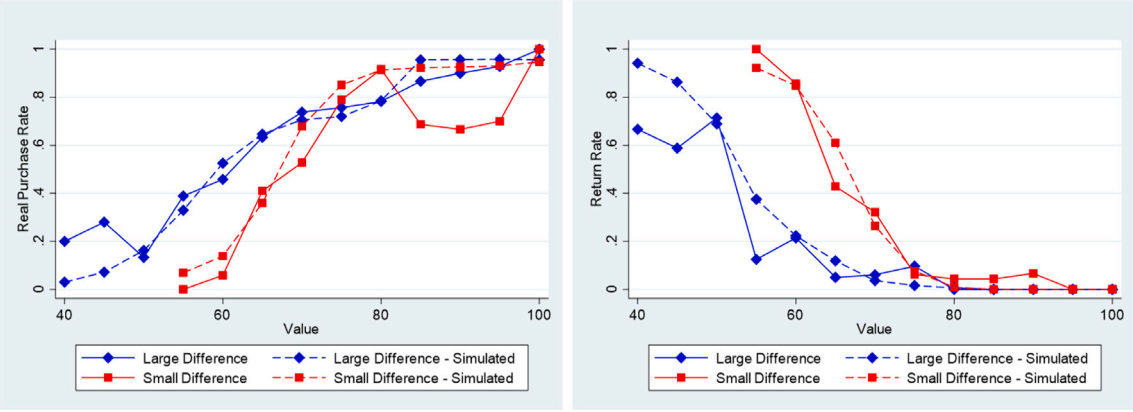


Fig. 2. Buyer's average real purchase rate and return rate for possible valuations in rounds 11–20.
The bin size of the valuation is 5 points.
The relationship between drawn valuations and real purchase rate is on the left.
The relationship between drawn valuations and return rate is on the right.

treatment reduces both purchase rate and real purchase rate, hurting the seller in addition to the relatively low uniform price. The large perception heterogeneity generates a high return rate, mainly reducing the return rate for high-type buyers while stimulating return behavior for low-type buyers, suggested by interaction effects shown in Table 7.

Result 3. *For both treatments, the presence of a return policy does not affect the purchase rate, but decreases the real purchase rate.*

Result 4. *There is no significant difference between the purchase rate across between-subjects treatments. However, the real purchase rate is significantly higher under the "Small Difference" treatment.*

Result 5. *The return rate is significantly higher under the "Large Difference" treatment.*

Table 8
OLS regressions on payoffs by roles.

	(1) Buyer	(2) Seller	(3) Buyer	(4) Seller
Large Difference	2.009	−5.961***	1.574	−6.873***
	(1.298)	(1.695)	(1.397)	(1.667)
With Return	0.228	−5.332**	0.228	−5.332**
	(1.938)	(2.359)	(1.930)	(2.348)
Large Difference × With Return	−0.240	2.684	−0.240	2.684
	(1.736)	(1.989)	(1.729)	(1.980)
Round	0.040	0.068	0.040	0.068
	(0.143)	(0.207)	(0.142)	(0.206)
Constant	61.653***	92.342***	65.019***	90.985***
	(1.418)	(2.473)	(1.262)	(1.687)
Survey Controls	Yes	Yes	No	No
Observations	1020	1020	1020	1020
R^2	0.026	0.038	0.004	0.024
Adjusted R^2	0.014	0.026	−0.000	0.020

Robust standard errors clustered at the subject level in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3. Profits

Finally, we compare the profit level by roles as the profit level of sellers (resp. buyers) represents the welfare level of producers (resp. consumers). The total payoff is significantly higher under the "Small Difference" treatment ($p \approx 0.02$)—the seller's profit is the driving force for such a high total payoff ($p < 0.01$), while the buyer's profit is roughly 2 points higher under the "Large Difference" treatment ($p \approx 0.07$).

Table 8 provides estimates of OLS regression results on the profit level by roles. The seller is worse off (1) under the "Large Difference" treatment and (2) when there is a return policy. Under the "Large Difference" treatment, the seller's profit decreases as more buyers benefit from the lower uniform price, and the seller faces two different layers of disadvantage that provide incentives to reduce the price: (1) the buyer will be less likely to purchase at a high price since he cannot return the good after purchase, and (2) the buyer's exact type is unknown to the seller, which increases the difficulty of screening different types of buyers. The interaction between the two channels has no significant effect on either role's payoffs, implying that the primary welfare impact lies in how each channel independently hurts sellers. Hence, we conclude that the direct effects of the two forces we examined negatively impact sellers but do not significantly increase buyers' payoffs.

Result 6. *Both the presence of a return policy and large perception heterogeneity reduce sellers' profits, but their impacts on buyers' profits are at best minimal.*

6. Conclusion

This paper aims to understand the direct effects of two distinct channels in sales platforms for experience goods: (1) the presence of a return policy and (2) heterogeneity in buyers' perceived valuation ranges. Limited attention has been paid to the case where the ranges of buyers' valuations can be different, which is an irreplaceable element in the context of experience goods. Furthermore, observational data seldom include people's perceptions of the experience good's valuation prior to purchase. As a result, a controlled environment appears to be an ideal setting for investigating these research questions.

Our results show that return policies generally do not increase the uniform price named by the seller in experience goods markets. Such an effect is strengthened by sellers facing more heterogeneous consumers and wanting to guarantee high-type buyers' purchases when there is no return policy. Accordingly, a lower uniform price, in addition to the large gap between consumers' potential valuations, both reduce the real purchase rate due to the additional option for buyers to return the good. Hence, the presence of a return policy or consumers' heterogeneous perceptions of the potential valuation range is likely to bring negative impacts, especially to sellers.

The main takeaway of the paper is that consumers' perception heterogeneity on valuation's range is a crucial aspect to keep in mind when policymakers predict or evaluate the direct effect of return policies, especially in online sales platforms with experience goods. After all, the direct effect of a return policy depends on how consumers perceive the range of the experience good's potential valuation, and the effect of mandated return policies can be negative. Our experimental investigation identifies the necessity to include consumers' perception heterogeneity in empirical research related to return policies.

Our study is subject to limitations. First, we only focus on mandated return policies. Future research can evaluate the potential effects of endogenous return policies—meaning that sellers can decide whether to offer a return policy to buyers. Second, we restrict

our attention to introducing the return policy to the market through within-subject variations. A natural next step would be to explore the effects of removing the return option. Third, sellers may avoid the cost of processing returns for low-value items by offering refunds without requiring the items to be sent back in practice, highlighting the need to compare return policies with alternative consumer protection measures. All three extensions will provide deeper insights into the flourishing literature of behavioral and experimental industrial organization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to proofread existing content. After using this tool/service, the authors reviewed and edited the content as needed. The authors take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jebo.2025.107287.

Data availability

Data will be made available on request.

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